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Adaptive Obstacle Avoidance Using Vision-Based Dynamic Prediction and Strategic Motion Planning

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ABSTRACT

In human–robot collaboration, ensuring both safety and efficiency in obstacle avoidance remains a critical challenge. This paper proposes a sampling-based danger-aware artificial potential field (SDAPF) method for obstacle avoidance during human–robot collaboration and interaction. Existing methods often struggle with dynamic obstacles and varying environmental complexities, which can hinder their performance. To address these challenges, SDAPF integrates three key components: position sampling for local minimum avoidance, a novel hazard index that quantifies risk based on the distance and relative velocity between the robot and dynamic obstacles and a dynamic obstacle motion prediction module leveraging depth image data. These features enable intelligent path selection, adaptive step size adjustments based on obstacle dynamics and proactive decision-making for collision-free navigation. The hazard index allows the robot to dynamically assess the urgency of avoiding an obstacle, whereas the motion prediction module anticipates future positions of moving obstacles, enabling the robot to plan paths in advance. The effectiveness of SDAPF is demonstrated through both simulations and real-world experiments, highlighting its potential to significantly enhance safety and operational efficiency in complex human–robot interaction scenarios.

1 | Introduction

Obstacle avoidance within the context of human–robot interaction is a subject of significant interest. As industrial advancements demand enhanced levels of collaboration between humans and robots, the imperative to plan paths that optimise both work efficiency and safety becomes paramount [1–3]. In addition to completing manipulation tasks, robots can also maximise their likelihood of success when collaborating with human users [4]. To achieve this, robots must employ the most efficient obstacle-avoidance paths, ensuring timely evasion of humans and thereby creating a secure environment for human–robot coexistence.

This paper mainly considers the safety issues of humans and the work efficiency of robots during human–robot interaction. We propose an obstacle avoidance algorithm based on an improved artificial potential field (APF) method, sampling-based danger-aware artificial potential field (SDAPF). Our contributions are as follows:

- We propose a novel velocity repulsion potential field function that incorporates the distance influencing factor and velocity influencing factor as risk determinants. The gradient descent of this potential field establishes the velocity repulsion function, facilitating collision-free path planning. Moreover, the robot intelligently assesses the relative speed and direction between itself and the human to determine optimal movement directions, obstacle avoidance strategies and paths. This adaptive decision-making enhances cooperative efficiency.
- By introducing an environmental complexity factor, we introduce a variable step size to our approach. Unlike

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conventional APF methods that rely on fixed step sizes, our approach dynamically adjusts step sizes based on environmental complexities. For instance, larger steps are appropriate when obstacles are distant or sparse, whereas smaller steps are necessary for closer obstacles. This adaptive step size approach aligns the algorithm with real-world scenarios and elevates planning efficiency.

- Inspired by the dynamic window method [5], we employ a sampling technique to evaluate positions in the subsequent moment. We analyse local minimum areas and risk indices for each sample point to identify the most secure and efficient option. By implementing this approach, we effectively evade local minima and derive solutions that are both safe and efficient.
- Inspired by the use of long short-term memory (LSTM) in autonomous driving to avoid pedestrians [6], we present a perceptual module and time-series model that accepts deep visual data, collects new human manipulation datasets and predicts the spatial location and motion trajectory of a human collaborator's hand movements. The system is allowed to perform computations in advance to avoid humans more efficiently and in a timely manner, ensuring the safety of human–robot interaction environments.

Visual localisation techniques employing 2D markers, as suggested in Garrido-Jurado et al. [7], facilitate the estimation of human hand movements. Alternative methods for gauging human motion include skeleton tracking [8] and 3D octree representations [9], both of which entail higher computational demands and increased noise levels. To illustrate our concepts, we employ the TIAGo manipulator, depicted in Figure 1, and assess them through a collaborative task that engages both human and robotic participants.

This paper improves and expands upon the algorithm presented in our previous publication [10]. In this updated version, the environmental complexity factor and variable step size are incorporated to enhance efficiency. Furthermore, the robot's risk level is quantified through a function that considers the speed and proximity between the robot and humans, facilitating smart decisions regarding advancing or retreating movements in response to their relative positions. We introduce a perception module based on depth image data and a time series prediction model to predict the hand motion of the human collaborator, allowing the robot to compute obstacle avoidance paths in advance based on these predictions.

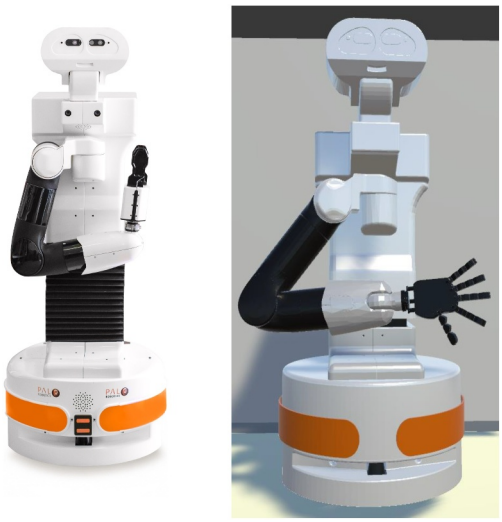


FIGURE 1 | TIAGo robot and its unified robot description format (URDF) model with base coordinates in the robot operating system (ROS).

In the experiment section, we supplement comparative experiments of SDAPF and Pivot [11] to demonstrate the advantages of our proposed algorithm in terms of efficiency and safety. These advantages are primarily reflected in path smoothness, path length and the success of planning at the same speed. Additionally, to demonstrate the effectiveness of our algorithm in complex environments, we compare the planning time and path length of our algorithm and rapidly exploring random tree (RRT) [12], illustrating the safety and efficiency of the proposed algorithm in such scenarios.

2 | Related Works

Recent years, significant advancements are seen in robot obstacle avoidance algorithms, each with its own set of strengths and challenges. For instance, the traditional APF [13], which is a local path planning algorithm, helps determine the subsequent movement scheme based on feedback motion information [14]. This is achieved by defining the repulsive force between the robot and the obstacle. However, it suffers from the drawback of easily getting trapped in a local minimum. A global path planning algorithm, proposed in Kavraki et al. [15], plans the entire path before execution of movement and is based on a sampling method. It constructs a growth graph using environment nodes obtained through sampling, when the manipulator's work environment is known. A global path is then identified through searching. Although this algorithm can identify the global optimum solution, it often requires extensive computational time [16].

Employing uniform sampling, the RRT [12] algorithm steers the growth of the path tree into the exploration zone, thereby charting a route from the origin to the destination. RRT can efficiently navigate multidimensional spaces with both convex and nonconvex obstacles [17]. Nonetheless, its weaknesses include slow search speed and insufficient real-time performance. To mitigate these issues, numerous RRT enhancements have been proposed and implemented in a variety of contexts. For example, Zhao et al. utilised the RRT-Connect algorithm for robotic arm path planning [18]. The RRT* algorithm, a renowned optimal motion planning algorithm [19]. Subsequently, Wen et al. applied an enhanced version of this approach, known as the dynamic domain fast search random tree algorithm (DDRRT), to unmanned aerial vehicle (UAV) online path planning [20].

Characterised by its computational agility and robustness, the APF method excels in facilitating real-time robot obstacle evasion, owing to its straightforward architecture and efficacy in timely responses. It is particularly esteemed in applications involving manipulator-based real-time obstacle navigation, where it constructs a repulsive potential around barriers. This is

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achieved by generating forces in the opposite direction of the potential field's gradient, ensuring a safe distance from obstacles [21]. An advanced version tailored for dynamic settings, which accounts for both trajectory and motion speed, is discussed [22]. Furthermore, in the context of cooperative human–robot tasks, a novel approach utilising APF to formulate a multijoint repulsion model that integrates depth sensory data with the robotic framework for enhanced collision avoidance is detailed [23].

However, APF's lack of global information often leads to stuck in a local minimum region where the repulsive force equals the attractive force, causing the target to become unreachable [24]. Fundamentally, APF centres on establishing a gravitational potential field at the target point and a repulsive potential field around obstacles. By descending along the negative gradients of these fields, gravitational and repulsive forces are engendered. In response to the vulnerability of APF to local minima, Yu et al. [25] counteracts this by introducing virtual obstacles, thereby preventing the robot from being trapped in genuine local minima. Meanwhile, Sun et al. [26] presimulates the locations of local minimum regions and strategically circumvents these during the planning process. The harnessing of the harmonic function was detailed in Ref. [27] as a mechanism to overcome the drawbacks tied to local minima.

A major advancement in this area of study was the introduction of the Pivot algorithm in Flacco et al. [11]. This innovative technique addresses scenarios where a manipulator is unable to retreat to evade an approaching obstacle moving at a velocity surpassing that of the manipulator. This algorithm, an enhancement of APF, dynamically alters the repulsion force vector's direction when confronted with rapidly approaching obstacles. It hinges on the principle of utilising APF-driven obstacle avoidance for slower obstacle speeds, transitioning to a modified repulsion force vector direction strategy when obstacles advance towards the robot arm at high velocities. Nevertheless, the mentioned paper did not offer a specific solution for cases where the repulsion vector and its variation are parallel.

Although the APF has demonstrated effectiveness in guiding robots around static obstacles, such as walls or objects, it proves inadequate in adequately addressing the dynamic and unpredictable nature of human hand movements [28]. Human hands are characterised by continuous motion, influenced by numerous factors, including task requirements and real-time interactions with the robot. Consequently, there is a pressing need for a more sophisticated approach. In recent studies, researchers have proposed innovative solutions to this challenge. Deep reinforcement learning (DRL) [29] is used to enable dynamic obstacle avoidance for robotic arms. The study leverages sensor inputs to learn how to avoid randomly moving obstacles, with experiments conducted primarily in simulated environments. DRL [30] is combined with reactive obstacle avoidance strategies, proposing a dynamic avoidance method for redundant manipulators, allowing the robot to continuously optimise its avoidance strategy through interactive learning. In Yu et al. [31], a coarse-to-fine framework is introduced that integrates global planning and local control to tackle the challenge of manipulating deformable linear objects (DLOs) in complex environments. This approach compensates for model inaccuracies, ensuring precise shape control and obstacle avoidance. Convolutional neural network-recurrent neural network (CNN-RNN) [32] models were employed to forecast human action types and force signals during manipulation. A safety-oriented framework for predicting hand movements using vision in tangible human–robot interaction settings was presented Wang et al. [33]. Further, Wu et al. [34] developed a system for forecasting gestures, leveraging the leap motion's capture of hand joint coordinates to anticipate dynamic gestures. The idea of action prediction has also been adopted in other recent works as demonstrated by Refs. [35–37].

3 | Methods and Principles

This section delineates our proposed framework and the underlying principles of our algorithm in detail. The algorithm unfolds as follows: initially, the next position of the robotic arm is sampled, followed by an assessment of the proximity to any obstacle. Should this proximity be within a safe margin, the velocity repulsion function is disregarded and path planning is conducted using the conventional APF approach, enriched by factors accounting for environmental complexity. Conversely, in instances where the proximity poses a threat, it becomes imperative to compute both the danger index and the velocity repulsion function. This adaptation, which incorporates a risk factor and a velocity repulsion mechanism into the established APF framework, enhances obstacle avoidance efficacy.

This chapter is deployed as follows. Initially, the sampling mechanism and how to use the sampling mechanism to avoid local minima are introduced. Then, the newly introduced danger index, including velocity impact factor and distance impact factor as well as the composition of the new velocity repulsive function, are introduced in detail. Lastly, the function and principle of environmental complex factors are introduced.

3.1 | Sampling-Based Motion Generator

APF has two shortcomings: one is easy to fall into local minimum value, the other is unreachable target. Our improvements proposed below can not only solve these two problems but also enable the robot to select the best obstacle avoidance direction according to the speed of obstacles.

To address the issue of potentially unreachable targets in APF, we incorporate the target's influence range into the calculation of repulsive potential energy, guaranteeing that the potential field attains its minimum exclusively at the target location. The new repulsive potential is as follows:

$$\boldsymbol{U}_{\mathrm{rep2}}(\boldsymbol{X}) = \begin{cases} \frac{k_{\mathrm{r}}}{2}\left(\frac{1}{p(\boldsymbol{X})} - \frac{1}{p_0}\right)^2 (\boldsymbol{X} - \boldsymbol{X}_{\mathrm{g}})^n & p(\boldsymbol{X}) \le p_0 \\ 0 & p(\boldsymbol{X}) > p_0 \end{cases} \quad (1)$$

where $\boldsymbol{X} - \boldsymbol{X}_{\mathrm{g}}$ represents the distance between the robot and the target and n is a constant that is greater than zero. The parameter k_{r} denotes the repulsion coefficient, which determines the magnitude of the repulsive force exerted by the

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obstacle on the robot. A higher value of k_r increases the intensity of this repulsion, so the robot can steer further away from obstacles. The function $p(\boldsymbol{X})$ represents the distance between the robot and the obstacle at position $\boldsymbol{X}$. When $p(\boldsymbol{X})$ is less than or equal to a predefined threshold p_0, the repulsive potential is activated, ensuring that the robot avoids the obstacle by generating a repulsive force. The improved repulsive force function is the negative gradient of the repulsive potential function:

$$\boldsymbol{F}_{\text{rep2}} = -n\frac{k_r}{2}\left(\frac{1}{p(\boldsymbol{X})} - \frac{1}{p_0}\right)^2 (\boldsymbol{X} - \boldsymbol{X}_g)^{n-1} \tag{2}$$

So, the final repulsion function is defined as follows:

$$\boldsymbol{F}_{\text{rep}}(\boldsymbol{X}) = \begin{cases} \boldsymbol{F}_{\text{rep1}}(\boldsymbol{X}) + \boldsymbol{F}_{\text{rep2}}(\boldsymbol{X}) & p(\boldsymbol{X}) \le p_0 \\ 0 & p(\boldsymbol{X}) > p_0 \end{cases} \tag{3}$$

where $\boldsymbol{F}_{\text{rep1}}$ is the repulsive force function of the traditional obstacle to the manipulator, and its direction is from the obstacle to the manipulator. $\boldsymbol{F}_{\text{rep2}}$ is the improved repulsive force function mentioned above, and the direction is from the target point to the manipulator. Where $\boldsymbol{F}_{\text{rep1}}$ and $\boldsymbol{F}_{\text{rep2}}$ are defined as follows:

$$\begin{cases} \boldsymbol{F}_{\text{rep1}} = k_r\left(\dfrac{1}{p(\boldsymbol{X})} - \dfrac{1}{p_0}\right)\dfrac{1}{p(\boldsymbol{X})^2}(\boldsymbol{X} - \boldsymbol{X}_g)^n \\ \boldsymbol{F}_{\text{rep2}} = -n\dfrac{k_r}{2}\left(\dfrac{1}{p(\boldsymbol{X})} - \dfrac{1}{p_0}\right)^2 (\boldsymbol{X} - \boldsymbol{X}_g)^{n-1} \end{cases} \tag{4}$$

To mitigate the local minima issue in robot navigation, this study introduces a novel approach termed SDAPF. This technique involves predicting multiple prospective positions for the subsequent time step, leveraging the robot's current coordinates (x_t, y_t, z_t) and the direction of the resultant force θ_t. Given the significant computational overhead associated with 3D space sampling, and our empirical observation that much of this space is superfluous, with the manipulator's optimal movement locus seldom falling within these regions, we propose optimising real-time efficacy by confining sampling to designated planes such as the yz, xy and xz planes. For illustrative purposes, consider the yz plane (refer to Figure 2): here, $p_{\min}$ denotes the effective radius for the distance impact factor within the APF framework, activating only when the manipulator-to-obstacle separation falls below a predefined threshold. Conversely, $p_{\max}$ represents the operational radius for the velocity impact factor in APF, with v_0 indicating obstacle velocity, $\Delta\theta$ the sampling angular increment and k_v the velocity's influence coefficient. The sampling orientations θ_w within the yz plane are delineated as follows:

$$\theta_w = \begin{cases} \theta_t + \text{sgn}(v_{oz})\, w\, \Delta\theta & k_v \le 0 \cap v_{oz} \ne 0 \\ \theta_t + w\, \Delta\theta & v_{oz} = 0 \\ \theta_t - \text{sgn}(v_{oz})\, w\, \Delta\theta & k_v > 0 \cap v_{oz} \ne 0 \end{cases} \tag{5}$$

Given that θ_t represents the projected resultant force direction on the robot in the yz plane, with θ_t spanning $(0...n)$, and v_{oz} denoting the obstacle's velocity component along the z-axis, we utilise the sign function sgn alongside the sample index $w = (0...n)$. The coefficients k_v and v_{oz} signify:

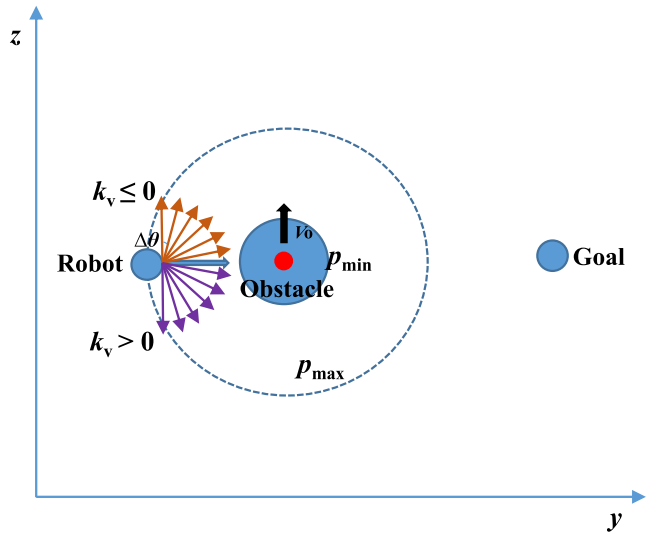


FIGURE 2 | Sampling-based APF for moving obstacle avoidance.

1. A positive k_v $(k_v > 0)$ signifies rapid obstacle motion, prompting the robot to sample future positions against the z-axis component v_{oz} of the obstacle's velocity, facilitating avoidance in scenarios where the obstacle's speed surpasses that of the robot.
2. A nonpositive k_v $(k_v \le 0)$ implies slow or negligible obstacle movement, leading the robot to preferentially sample in the direction aligned with v_{oz}, enabling faster circumvention in the forward direction.
3. In cases of stationary obstacles or when $v_{oz} = 0$, the robot opts for sampling in the positive direction of θ_t, effectively navigating over the obstacle. The enumeration of potential future positions for the robot's next move is calculated as follows:

$$\begin{bmatrix} x_0 & y_0 & z_0 \\ \cdots & \cdots & \cdots \\ x_n & y_n & z_n \end{bmatrix} = \begin{bmatrix} x_t & y_t + d\cdot\cos\theta_0 & z_t + d\cdot\cos\theta_0 \\ \cdots & \cdots & \cdots \\ x_t & y_t + d\cdot\cos\theta_n & z_t + d\cdot\cos\theta_n \end{bmatrix} \tag{6}$$

where the set of coordinates $(x_0...x_n, y_0...y_n, z_0...z_n)$ delineates the potential positions evaluated during simulation, with d denoting the incremental step size for sampling. The selection of the optimal movement direction is governed by an evaluation function, which is articulated as follows:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \min\left(\boldsymbol{P}, \boldsymbol{F}\left(\begin{bmatrix} x_0 & y_0 & z_0 \\ \cdots & \cdots & \cdots \\ x_n & y_n & z_n \end{bmatrix}\right)\right) \tag{7}$$

where $\boldsymbol{P}$ represents the set of possible positions (x, y, z) that the robot can move to in the next step. The function $\boldsymbol{F}$ is the evaluation function that calculates the total magnitude of forces acting on the robot at each of these positions, including the attractive, repulsive and velocity-based forces. The evaluation process aims to identify the position that minimises $\boldsymbol{F}$, thereby selecting the optimal movement direction that ensures efficient and safe obstacle avoidance.

The determined position for the subsequent moment minimises the total magnitude of forces acting on the robot. Upon the

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manipulator's end effector approaching the obstacle's influence zone, the SDAPF technique simulates potential positions, calculating the force for each scenario. This approach allows the robot to proactively predict its next move, thereby circumventing local minima. Moreover, the evaluation incorporates both the direction and magnitude of the relative velocity between the robot and the obstacle, facilitating an optimal avoidance path that balances efficiency and safety.

3.2 | Danger Index for Safety and Efficiency

In the interest of human safety and operational efficiency, the key determinants of the danger index are the relative position and speed between the robot and human, delineated as follows:

1. The distance impact factor is defined as follows:

$$f_d = \begin{cases} \eta\left(\dfrac{1}{p(\boldsymbol{X})} - \dfrac{1}{p_{\max}}\right) & p(\boldsymbol{X}) \le p_{\max} \\ 0 & p(\boldsymbol{X}) > p_{\max} \end{cases} \tag{8}$$

where $\eta = p_{\max} \cdot p_{\min} / \left(p_{\max} - p_{\min}\right)$ is the distance scale factor and $p_{\max}$ and $p_{\min}$ represent the maximum and minimum influence radius of the moving obstacle, respectively. The distance impact factor is affected by the distance relationship between the end effector and the dynamic obstacle, for example, the distance impact factor is greater when the robot is closer to the obstacle.

2. The velocity impact factor is defined as follows:

$$k_{\mathrm{v}} = \operatorname{sgn}(\gamma|\boldsymbol{v}_{\mathrm{o}}| - |\boldsymbol{v}_{\mathrm{r}}|) \tag{9}$$

where γ represents the velocity scaling coefficient, with $|\boldsymbol{v}_{\mathrm{o}}|$ and $|\boldsymbol{v}_{\mathrm{r}}|$ denoting the absolute velocities of the dynamic obstacle and the robot's end effector, respectively. The function sgn() indicates the signum operation. A positive velocity impact factor emerges when the dynamic obstacle's velocity, scaled by γ, exceeds that of the end effector, and conversely, it results in a negative value. In this paper, a uniform motion model is adopted for the obstacle.

The specific velocity repulsive potential field function is defined as a combination of the distance impact factor and the velocity impact factor, which is as follows:

$$\boldsymbol{U}_{\mathrm{rep}}(\boldsymbol{v}) = \begin{cases} \dfrac{k_{\mathrm{ro}} f_d \boldsymbol{v}^2}{2} & f_d > 0 \cap \alpha \in \left(-\dfrac{1}{2}, \dfrac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

where v represents the relative velocity between the end effector and the obstacle. It is defined as follows, note that the speeds here are all vectors and vector operations are performed. When $k_{\mathrm{v}} > 0$, the speed of the obstacle is greater than the speed of the robot and it is safer for the robot to move in the opposite direction of the obstacle movement, so the combined speed is the vector sum of the robot speed and the opposite direction of the speed of the obstacle v_{o}. When $k_{\mathrm{v}} < 0$, the speed of the obstacle is less than the speed of the robot and the robot can avoid the obstacle at a faster speed, so the combined speed is the vector sum of the speed of the robot and the speed of the obstacle in the positive direction. Here, k_{ro} is the repulsive gain coefficient and α is the normalised angle determining frontal approach, activating the potential only when collision risk is high.

$$\boldsymbol{v} = \begin{cases} \boldsymbol{v}_{\mathrm{r}} - \boldsymbol{v}_{\mathrm{o}} & k_{\mathrm{v}} \le 0 \\ \boldsymbol{v}_{\mathrm{r}} + \boldsymbol{v}_{\mathrm{o}} & k_{\mathrm{v}} > 0 \end{cases} \tag{11}$$

$\boldsymbol{U}_{\mathrm{rev}}$ represents the velocity potential energy and k_{ro} is the scale factor. The angle α, formed between the relative velocity vector v and the vector depicting the relative position from the robot's end effector to the obstacle, serves to ascertain the robot's directional movement relative to the obstacle. The inclusion of the velocity impact factor enables the formulation of nuanced avoidance tactics in response to the varying kinematics of mobile obstacles. Concurrently, the distance impact factor permits the modulation of velocity-based repulsion in alignment with the fluctuating proximity between the end effector and the obstacle.

The derivation of the velocity repulsion force is accomplished by taking the negative gradient of the velocity repulsive potential, which is articulated as follows:

$$\boldsymbol{F}_{\mathrm{vel}} = \begin{cases} -k_{\mathrm{ro}} f_d (\boldsymbol{v}_{\mathrm{r}} - \boldsymbol{v}_{\mathrm{o}}) & k_{\mathrm{v}} \le 0 \cap f_d > 0 \cap \alpha \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \\ -k_{\mathrm{ro}} f_d (\boldsymbol{v}_{\mathrm{r}} + \boldsymbol{v}_{\mathrm{o}}) & k_{\mathrm{v}} > 0 \cap f_d > 0 \cap \alpha \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

The activation of the velocity repulsion force, denoted as $\boldsymbol{F}_{\mathrm{vel}}$, occurs when the proximity between the robot's end effector and an obstacle falls below the threshold $p_{\max}$. This repulsion force $\boldsymbol{F}_{\mathrm{vel}}$ is directed antithetically to the velocity vector $\boldsymbol{v}$. To augment path planning within the SDAPF framework, the velocity potential field $\boldsymbol{U}_{\mathrm{rev}}$ is integrated. The composite force field influencing the robot's end effector is characterised as follows:

$$\boldsymbol{U} = \boldsymbol{U}_{\mathrm{attr}} + \boldsymbol{U}_{\mathrm{rep}} + \boldsymbol{U}_{\mathrm{rev}} \tag{13}$$

3. The environmental complexity factor is defined as follows:

$$\beta = \begin{cases} 0.5 & |\theta_t - \theta_{t-1}| < \dfrac{\pi}{9} \\ 1.0 & \dfrac{\pi}{9} \le |\theta_t - \theta_{t-1}| < \dfrac{\pi}{4} \\ 2.0 & \dfrac{\pi}{4} \le |\theta_t - \theta_{t-1}| \end{cases} \tag{14}$$

where θ_t and $\theta_{(t-1)}$ represent the direction of the resultant force at t and t_1, respectively, and the complexity of the environment can be analysed by the change of the angle between them: the larger the angle between the resultant forces at t and t_1, the more complex the environment is, and the higher the environmental complexity factor β should be, so the step size should be reduced. In contrast, the smaller the angle between the two forces is, the smaller

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the environmental complexity factor β is and the more the step size should be increased, in inverse proportion. Based on the environmental complexity factor, the step size function is defined as follows: β thus acts as an adaptive gain that modulates step size according to directional instability of the resultant force, reflecting local navigational complexity.

$$s = \frac{s_0}{\beta} \tag{15}$$

where s_0 represents the initial step size. In this way, when the manipulator is close to the obstacle, to avoid the obstacle successfully, the step size should be small, and when the manipulator is far from the obstacle, the step size becomes larger, which can ensure the safety and efficiency of human–robot interaction. The experiments show that the efficiency of the variable-step-size path planning algorithm is higher than that of the fixed-step-size path planning algorithm.

3.3 | Obstacle Trajectory Prediction

The goal of our proposed obstacle motion trajectory prediction module is to accept depth image data and predict the spatial position and motion trajectory of a human collaborator's hand motion from visual inputs. In this setup, the sole input consists of video frames obtained from a camera positioned atop the robot. For the sake of simplicity, it is initially presumed that the human collaborator manipulates a single object within the workspace using one hand. The video frame captures the hand motion trajectory of the human collaborator.

To encode hand movements from the present frame to the subsequent time step, we utilise pixel-level displacement measurements ($\Delta T, \Delta \boldsymbol{X}, \Delta \boldsymbol{Y}, \Delta \boldsymbol{Z}$). Addressing the intricacies presented by complex and dynamic obstacle motion, we introduce a sophisticated augmentation to our trajectory prediction module through the incorporation of a multilayer bidirectional LSTM network. This augmentation is meticulously designed to capture intricate temporal patterns more effectively than traditional LSTM models. We depict the overall module in Figure 3 and describe different components in detail.

A distinctive attribute of the multilayer bidirectional LSTM model lies in its capability to assimilate information from both past and future temporal segments. In direct contrast to the conventional unidirectional LSTMs that focus solely on historical data, the bidirectional nature of our model empowers it to harness forthcoming context, thus bolstering its comprehension of obstacle motion dynamics. To make this paper self-contained, we briefly introduce the multilayer bidirectional LSTM model here.

Mathematically, let us denote the historical sequence of obstacle positions as $\boldsymbol{X} = \{(\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \boldsymbol{x}_t)\}$, where $\boldsymbol{x}_t$ denotes the obstacle position at time step t. The multilayer bidirectional LSTM processes this sequence in both forward and reverse directions, resulting in a sequence of hidden states $\boldsymbol{H} = \{(\boldsymbol{h}_1, \boldsymbol{h}_2, ..., \boldsymbol{h}_t)\}$. To

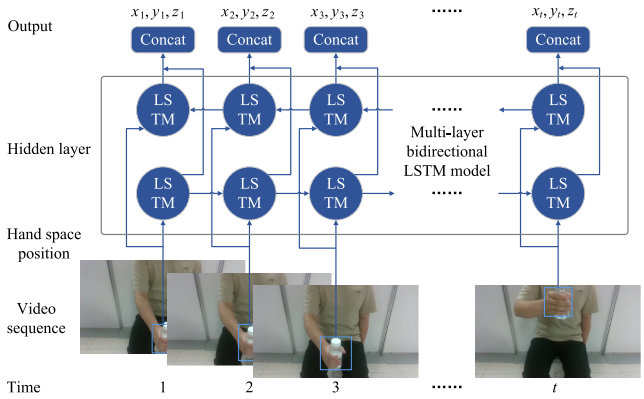


FIGURE 3 | Sampling-based APF for moving obstacle avoidance.

formalise the operation of the multilayer bidirectional LSTM, the mathematical formulation is as follows:

$$\boldsymbol{h}_t = \mathrm{LSTM}^{\rightarrow}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}) \tag{16}$$

$$\boldsymbol{h}_t = \mathrm{LSTM}^{\leftarrow}(\boldsymbol{x}_t, \boldsymbol{h}_{t+1}) \tag{17}$$

$$\boldsymbol{h}_t = [\boldsymbol{h}_t^{\rightarrow}, \boldsymbol{h}_t^{\leftarrow}] \tag{18}$$

where $\boldsymbol{h}_t^{\rightarrow} = \mathrm{LSTM}^{\rightarrow}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}^{\rightarrow})$ denotes the forward LSTM pass, calculating the hidden state $\boldsymbol{h}_t^{\rightarrow}$ at time t based on the current obstacle position $\boldsymbol{x}_t$ and the previous hidden state $\boldsymbol{h}_{t-1}^{\rightarrow}$. Conversely, $\boldsymbol{h}_t^{\leftarrow} = \mathrm{LSTM}^{\leftarrow}(\boldsymbol{x}_t, \boldsymbol{h}_{t+1}^{\leftarrow})$ represents the backward LSTM pass, where $\boldsymbol{h}_t^{\leftarrow}$ is computed using $\boldsymbol{x}_t$ and the subsequent hidden state $\boldsymbol{h}_{t+1}^{\leftarrow}$. The overall hidden state at time t, $\boldsymbol{h}_t = [\boldsymbol{h}_t^{\rightarrow}, \boldsymbol{h}_t^{\leftarrow}]$, is then obtained by combining these forward and backward hidden states, thus encapsulating both past and future information at this point in the sequence. This methodology enables the bidirectional LSTM to effectively capture complex temporal dynamics within the data.

By amalgamating insights from both past and future context, our model excels in capturing intricate patterns that might be overlooked by conventional LSTMs. This enhanced contextual understanding results in more precise trajectory predictions, particularly when confronted with swiftly evolving obstacle behaviours.

Moreover, the multilayer architecture of our model enables it to capture hierarchical representations of obstacle motion. Each layer acquires diverse levels of abstraction, working together to provide a more detailed understanding of underlying motion patterns.

Through empirical evaluations, we empirically substantiate the superiority of the multilayer bidirectional LSTM over conventional LSTMs in accurately predicting obstacle trajectories.

To quantify the performance of our model in predicting obstacle trajectories, a mathematical formulation is employed as shown below. The predicted trajectory at each time step t, denoted as $\boldsymbol{y}_t$, is computed using a linear function of the hidden state $\boldsymbol{h}_t$ obtained from the LSTM. The accuracy of these predictions is evaluated using a mean squared error loss function $\mathcal{L}$, which

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measures the difference between the predicted trajectory $\boldsymbol{y}_t$ and the true trajectory $\boldsymbol{y}_{\text{true},t}$. This is formulated as follows:

$$\boldsymbol{y}_t \underset{T}{=} \text{Linear}(\boldsymbol{h}_t) \tag{19}$$

$$\mathcal{L} = \sum_{t=1} \|\boldsymbol{y}_t - \boldsymbol{y}_{\text{true},t}\|^2 \tag{20}$$

This enhancement translates into elevated reliability and effectiveness in devising obstacle avoidance strategies, thus further augmenting the safety and efficiency of human–robot interaction.

The multilayer bidirectional LSTM model is trained using historical motion data related to obstacles. This dataset comprises discrete time-step positions of obstacles. Each time step is characterised by a vector of coordinates ($T, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z}$) signifying the obstacle's position in three-dimensional space. The multilayer bidirectional LSTM model processes a sequence of historical data points as input, calculated pixel-level displacement ($\Delta T, \Delta \boldsymbol{X}, \Delta \boldsymbol{Y}, \Delta \boldsymbol{Z}$) and learns to predict forthcoming positions of obstacles.

The training data are structured as follows: each historical position sequence is employed to predict the positions of the obstacle for several ensuing time steps. This sequence-to-sequence prediction mechanism enables the multilayer bidirectional LSTM model to capture temporal correlations and internalise underlying motion patterns.

Throughout the training process, the model is optimised to minimise the mean squared error between the predicted positions and the actual positions of obstacles in subsequent time steps. Upon training, the multilayer bidirectional LSTM model generalises its learnt patterns, facilitating real-time prediction of obstacle trajectories.

In the deployment phase, the trained multilayer bidirectional LSTM model is deployed to predict obstacle trajectories. Given the historical positions of an obstacle, the LSTM model generates predictions for its forthcoming positions. These projections offer invaluable insights into the obstacle's future trajectory, equipping our system to anticipate and formulate avoidance strategies.

Subsequently, the projected obstacle trajectories are incorporated into our path planning algorithm. By simultaneously considering the anticipated obstacle positions and the robot's motion, our system devises safer and more efficient routes to navigate around obstacles. The predictions of obstacle trajectories enhance the agility and adaptability of the robot's path planning within dynamic environments.

4 | Experiments

4.1 | Experimental Setup

Our framework's performance was assessed using a TIAGo robot equipped with a 7-DOF (Degree-of-Freedom) arm capable of redundant movements. The focus of this evaluation was on a typical human–robot interaction scenario where the robot's end effector and the human hand move in orthogonal directions, specifically the end effector to the right and the human hand upward. Priority is given to safety, balanced with the need for maintaining robotic efficiency, necessitating intelligent control decisions by the robot in response to the human hand's movements. Human hand motion is tracked via a 2D ArUco Marker affixed to an object held by the hand. While experiments included skeleton-based tracking with RGB-D sensors like Kinect, the variability in human joint point detection led to unreliable hand velocity estimates. The 2D ArUco Marker proved to be a more reliable and swift solution for our assessments. The location of the object held by the human is converted into the robot's coordinate frame, as illustrated in Figure 4 [38]. The SDAPF algorithm developed herein facilitates the computation of a collision-free trajectory for the manipulator's end effector, which informs the robotic control via inverse kinematics [39]. Experiment parameters are compiled in Table 1. The computations were performed on a remote computer with the following configuration: Intel Core i7-10750H CPU, 32 GB RAM, and an NVIDIA RTX 2060 6 GB GPU. The robot and the computer are connected via a network cable. The computation time was nearly real-time, with an average delay of 235 ms and a maximum delay of 307 ms across all experiments.

To demonstrate the proposed idea, we perform comparative simulation experiments with the Pivot algorithm [40] on the length and smoothness of the obstacle avoidance path. Because our method and Pivot algorithm are improvements of the traditional APF method, the experimental results can fully prove the advantages and disadvantages of the two improved methods.

4.2 | Dataset

To validate the previously proposed hand motion prediction module, a testing platform was established to facilitate diverse users engaging in various manipulation actions with multiple objects. To mitigate occlusion, the Intel RealSense D435i Depth camera was deployed in real-world scenarios at a fixed height equivalent to that of the robot's body camera. This setup captured depth video data, and an action capture system leveraged YOLOv5 to identify the actual hand positions of users in world coordinates during the execution of manipulation actions. These

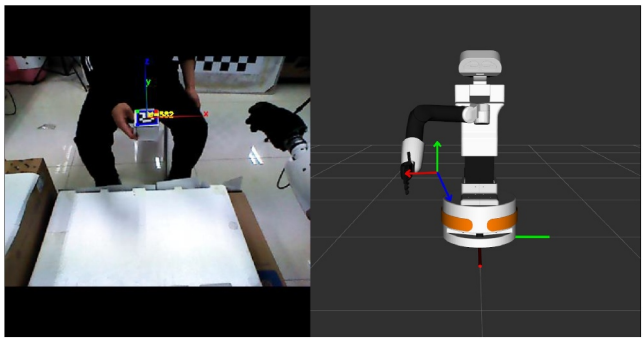


FIGURE 4 | A vision-based real-time estimation of the motion status of a human hand using the ArUco Marker.

TABLE 1 | The experiment parameters.

p_0	k_a	k_r	n	$\Delta\theta$	p_{max}	p_{min}	k_{ro}	γ
0.1 m	1	1	15	6°	0.2 m	0.1 m	10	1

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positions were recorded in the format $(T, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z})$ and compiled into a dataset.

Departing from real-world human–robot collaborative scenarios, three distinct actions were designed, each associated with a target object (shown in Figure 5 and Table 2). For each action-object pair, human participants were directed to perform the same action five times. Consequently, a dataset comprising 9000 frames was collected from this testing platform, which was used as the training data for the obstacle trajectory prediction model.

To demonstrate the performance of our motion prediction module, we show an example of the prediction output in Figure 6, where Figure 6a,b plots the historical spatial position of the hand and the prediction results in a 3D image. Figure 6c depicts the predicted results of the hand motion at a particular frame.

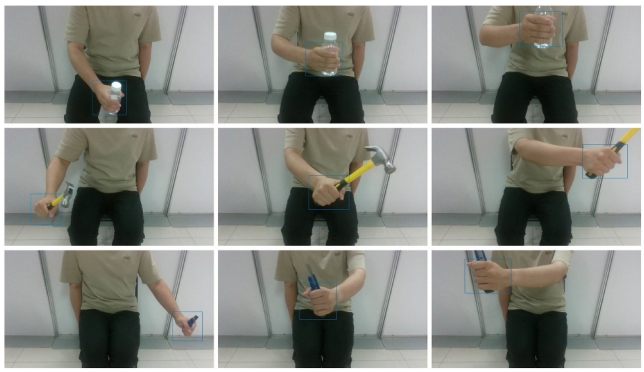


FIGURE 5 | Some examples of the dataset.

TABLE 2 | Object-action pairs in the dataset.

Object	Action
Bottle	Pick up, drink water
Hammer	Lift, pound
Knife	Lift, brandish

4.3 | Experimental Results

4.3.1 | Simulation Scenario 1: Static Obstacle

For the first experiment, a cube-shaped obstacle remains stationary, positioned along the path from the robot's end effector to the target. This obstacle is represented as a cylindrical volume, expanded to account for the dimensions of the end effector and a predefined safety margin. The robot employs an ArUco marker to identify and locate this cube obstacle. Since the cube obstacle is static, the prediction result of the obstacle trajectory prediction module is the current position of the obstacle and the obstacle avoidance process is shown in Figure 7. Figure 7a,c depicts the obstacle avoidance process using the SDAPF and Pivot methods in simulation scenarios. The recorded trajectories (blue), obtained by the SDAPF, APF and Pivot algorithms, are displayed in Figure 7d–f, respectively. Figure 7g shows the spikes and path lengths for the paths generated by the SDAPF and Pivot algorithms during obstacle avoidance.

The experimental findings demonstrate that the manipulator is capable of real-time detection of the static cube obstacle's location and effectively circumvents it using both SDAPF and Pivot methods, whereas the traditional APF algorithm fails during computation and collides with the obstacle. However, as clearly shown in Figure 7d,f,g, the Pivot algorithm generates a longer path with more spikes throughout the obstacle avoidance process in the same experimental scenario. This observation is further reflected in Tables 3 and 4. In contrast, the proposed SDAPF algorithm, through position sampling and an evaluation function, obtains a smooth avoidance path along the edge of the obstacle's influence radius on the same plane.

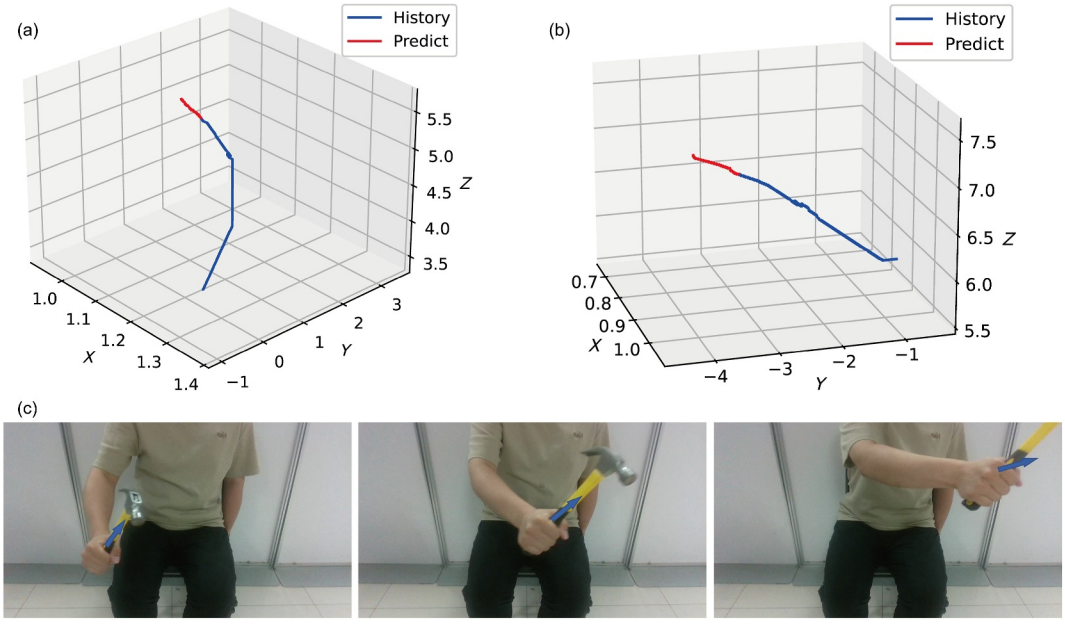


FIGURE 6 | Examples of hand motion prediction results. Panels (a and b) are 3D path images depicting the historical trajectory and prediction results, and the direction and length of the blue arrow superimposed on the hand in panel (c) illustrate the predicted hand motion at a specific frame.

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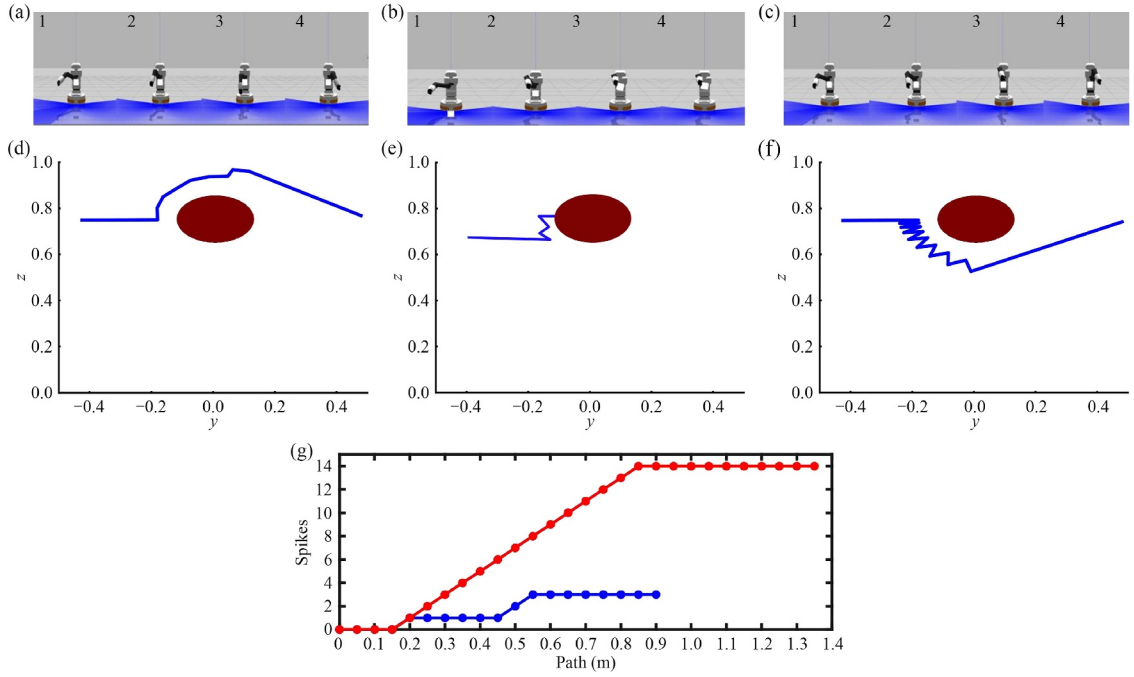


FIGURE 7 | The comparative experimental results using the proposed SDAPF algorithm, APF algorithm and Pivot algorithm for avoiding a static human hand are as follows: Panels (a–c) are snapshots of the collision avoidance process using the SDAPF, APF and Pivot algorithms, respectively. Panels (d–f) show the complete obstacle (red) avoidance paths (blue) of the manipulator in a 2D view using the SDAPF, APF, and Pivot methods, respectively. Panel (g) presents the statistics of the number of spikes and the path lengths from the path diagrams in panels (d) and (f), where the red line represents the processing result of the SDAPF algorithm, and the blue line represents the processing result of the Pivot algorithm. The horizontal axis indicates the path length, and the vertical axis represents the number of spikes.

In further experiments, we adjusted the size and quantity of the obstacles to evaluate the performance of the algorithms in different scenarios. First, the size of the obstacle was doubled, meaning the length, width and height were all increased by a factor of two as shown in Figure 8a. Second, the number of obstacles was changed to three, arranged horizontally, as shown in Figure 8b. Finally, the number of obstacles was set to three, arranged vertically, as shown in Figure 8c. The trajectories (blue), obtained by the SDAPF and Pivot algorithms, are displayed in Figure 8d–i.

The experimental results indicate that, in the scenario where the obstacle's size was increased, the SDAPF algorithm still managed to generate smooth and stable avoidance paths, whereas the Pivot algorithm resulted in longer paths with more spikes. When the number of obstacles was increased to three, whether arranged horizontally or vertically, the SDAPF algorithm continued to demonstrate stable obstacle avoidance performance. However, in the scenario with three vertically arranged obstacles, the Pivot algorithm failed to successfully avoid the obstacles, further highlighting its limitations in more complex environments. These experiments further verify the superiority of the SDAPF algorithm in handling various obstacle scenarios.

4.3.2 | Simulation Scenario 2: Crossover With Slow Speed

In this scenario, to clearly show that our algorithm has the advantage of intelligently determining the direction of obstacle avoidance, we make the obstacle move along the z-axis. The cube-shaped obstacle traverses the robot's designated trajectory at a consistent slow velocity, which remains below the robot's speed. The avoidance manoeuvre utilises parameters such as $V_{rob} = 0.1$ m/s for the robot and $V_{obs} = 0.08$ m/s for the obstacle. The robot estimates the position and velocity of the cube obstacle by detecting the ArUco. Since the trajectories of cubic obstacles are relatively slow and uniform, the predictions of the obstacle trajectory prediction module are similar to the results of a linear fitting with a visual camera. According to our proposed algorithm, the velocity impact factor and velocity repulsion vector are calculated, which can drive the end effector to move ahead of the obstacle. Similarly, we use the Pivot algorithm for comparison experiments in the same scenario. The experimental results are shown in Figure 9.

TABLE 3 | Comparison results of our method and the Pivot method for the avoidance route.

V_{obs} (m/s)	0	0.06	0.08	0.12	0.14
SDAPF (mm)	900	850	1000	900	850
Pivot (mm)	1350	1060	1050	Fail	Fail

TABLE 4 | Comparison results of our method and the Pivot method in terms of the number of spikes.

V_{obs} (m/s)	0	0.06	0.08	0.12	0.14
SDAPF (mm)	3	3	3	2	850
Pivot (mm)	14	4	6	Fail	Fail

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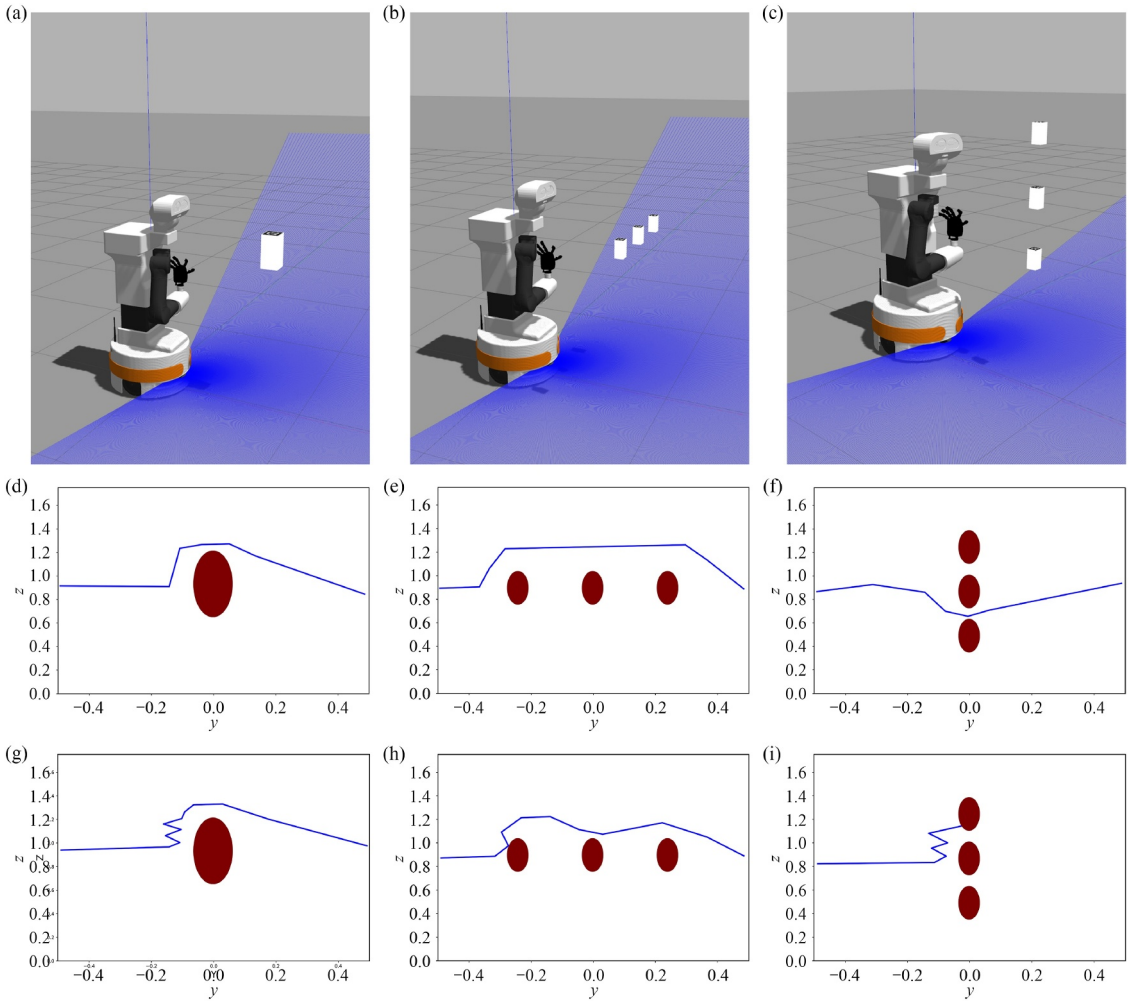


FIGURE 8 | Comparative experimental results using the proposed SDAPF algorithm and Pivot algorithm for avoiding obstacles in different scenarios: Panels (a–c) represent the scenarios where the robot faces an enlarged obstacle, three horizontally arranged obstacles and three vertically arranged obstacles, respectively. Panels (d–i) show the complete obstacle (red) avoidance paths (blue) of the manipulator in a 2D view using the SDAPF and Pivot algorithms.

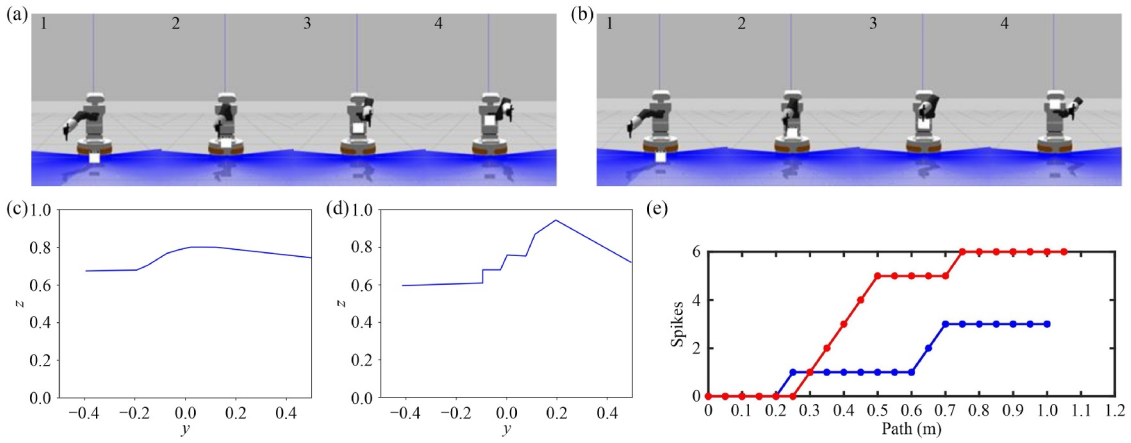


FIGURE 9 | Comparative experimental results using the proposed SDAPF algorithm and Pivot algorithm for avoiding a slow moving cube. The cube moves upward along the z-axis at a nearly uniform speed. Panels (a and b) are snapshots of the collision avoidance process using SDAPF and Pivot, respectively, panels (c and d) show the complete obstacle avoidance path (blue) of the manipulator in a 2D view by using SDAPF and the Pivot approach, respectively, and panel (e) shows the statistics of the numbers of spikes and the path lengths for the two path diagrams in panels (c and d); the red line represents the processing result of the SDAPF algorithm, the blue line represents the processing result of the Pivot algorithm, the horizontal axis is the path length and the vertical axis is the number of spikes.

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As shown in Figure 9, both the SDAPF and Pivot algorithms can successfully overcome obstacles. However, The Pivot algorithm is not as good as the SDAPF algorithm in terms of path length and smoothness due to the lack of evaluation function constraints and position sampling strategies as shown in Tables 3 and 4.

For the purpose of this study, slow and fast obstacle speeds are defined relative to typical human motion patterns. A slow-moving obstacle refers to one with a velocity of approximately 0.08 m/s, representing deliberate hand movements during collaborative tasks. A fast-moving obstacle is defined as having a velocity of 0.12 m/s, simulating quicker gestures or sudden motions that require more immediate avoidance behaviour.

4.3.3 | Simulation Scenario 3: Crossover With Fast Speed

Here, we consider the manipulator crossing a fast-moving cube, which is not moving in a straight line, but rather following a trajectory similar to the manoeuvring path of a real-life collaborator and at a speed greater than that of the manipulator as shown in Figure 10. The parameters used for the avoidance action are $V_{rob} = 0.1$ m/s and $V_{obs} = 0.12$ m/s. Under these circumstances, the system determines that navigating behind the obstacle is the safer course of action for the robot based on the anticipated path of the obstacle derived from the trajectory prediction module. This objective can be achieved through the implementation of the SDAPF algorithm, which introduces innovative definitions of potential functions that account for the relative velocities involved.

We conducted an ablation study by comparing the performance of the SDAPF algorithm with two variants: the traditional APF algorithm enhanced with a multilayer bidirectional LSTM network for obstacle trajectory prediction and the Pivot algorithm. In contrast, the Pivot algorithm opts for avoidance in the positive direction, which ultimately leads to a collision. The red cross in Figure 10f indicates that the manipulator collided with the obstacle when using the Pivot algorithm, and the path was

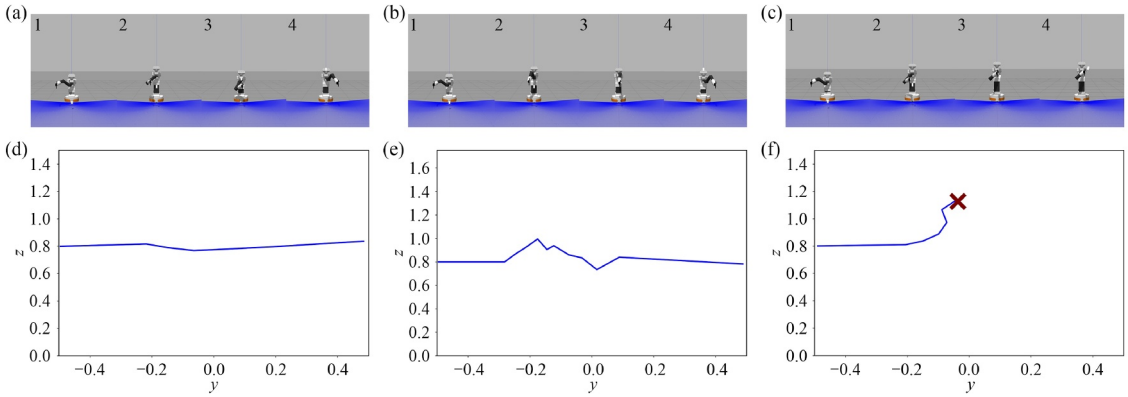


FIGURE 10 | Comparative experimental results using the proposed SDAPF algorithm, the APF algorithm with a prediction network and the Pivot algorithm for avoiding a fast-moving cube. The cube moves upward along the z-axis at a nearly uniform speed. Panels (a–c) represent snapshots of the collision avoidance process using the SDAPF algorithm, the APF algorithm with a prediction network and the Pivot algorithm, respectively. Panels (d–f) represent the complete obstacle avoidance path (blue) of the manipulator in a 2D view and the positions where planning failed after a collision with the obstacle (red).

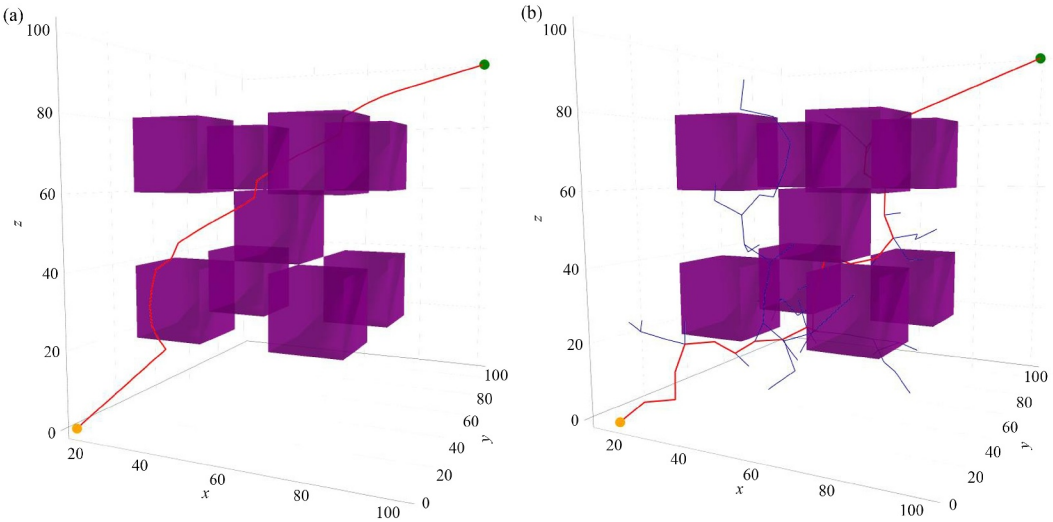


FIGURE 11 | Comparison of the experimental results of using the (a) SDAPF algorithm and (b) RRT algorithm to avoid obstacles in a static multiobstacle environment.

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not drawn after the collision. The SDAPF algorithm with trajectory prediction effectively forecasts the obstacle's path in advance, determining that the obstacle will not impede the forward direction, thus resulting in a smoother trajectory. In comparison, the traditional APF algorithm lacks a position sampling mechanism, leading to more abrupt changes in the path. The experimental results also demonstrate the effectiveness of the SDAPF algorithm.

To validate the effectiveness and safety of our approach, we conducted repeated experiments in various scenarios where the manipulator encountered obstacles in different motion states. As shown in Tables 3 and 4, the proposed SDAPF algorithm significantly enhances collision avoidance during crossover tasks. Moreover, the introduced framework seamlessly integrates considerations for both safety and efficiency. The robotic system exhibits the capability to make intelligent decisions based on the dynamic status of the obstacle, such as determining whether to advance ahead of or trail behind the moving hand. This feature distinguishes it from conventional obstacle avoidance methodologies.

4.3.4 | Simulation Scenario 4: Complex Scenarios

To verify the applicability of our proposed SDAPF algorithm in multiobstacle and complex scenarios, we compare the application effect of the SDAPF algorithm in a multiobstacle environment with the effect of the RRT algorithm, which has obvious advantages in path planning in high-dimensional and complex scenarios compared with other algorithms. We placed multiple obstacles along the path between the robot and the target point, and the experimental scene is shown in Figure 11. The path planned by the proposed SDAPF algorithm is shown in Figure 11a, and the path planned using the RRT algorithm in the same scenario is shown in Figure 11b. Table 5 shows the planning time and path length obtained by the two path planning algorithms. From the table, we can see that the SDAPF algorithm we proposed is better than the RRT algorithm in terms of planning time and path length. Comparing Figure 11a,b, it can be seen that the path planned by our method is smoother, more efficient and more effective and that it lacks shortcomings such as the nonoptimal chosen path and high randomness of the RRT algorithm.

TABLE 5 | Comparison results of our SDAPF and RRT.

Algorithm	Planning time (s)	Path length (cm)
SDAPF	0.071	185
RRT	0.094	217

4.3.5 | Real Scene Experiment

In this section, we verify the experimental results of the simulation experiments in a real scene. In this experiment of collision avoidance, the workspace of the manipulator is monitored by a depth sensor, which is located on the head of the robot. The parameters used in the real scene are the same as those used in the simulation experiment.

As depicted in Figure 12, the SDAPF algorithm's performance aligns with the simulation results for the crossover task. Moreover, the introduced framework meticulously addresses both safety and efficiency considerations. The robotic system exhibits intelligent decision-making capabilities based on the human's motion status, determining whether to advance ahead of or trail behind the moving hand. This characteristic distinguishes it

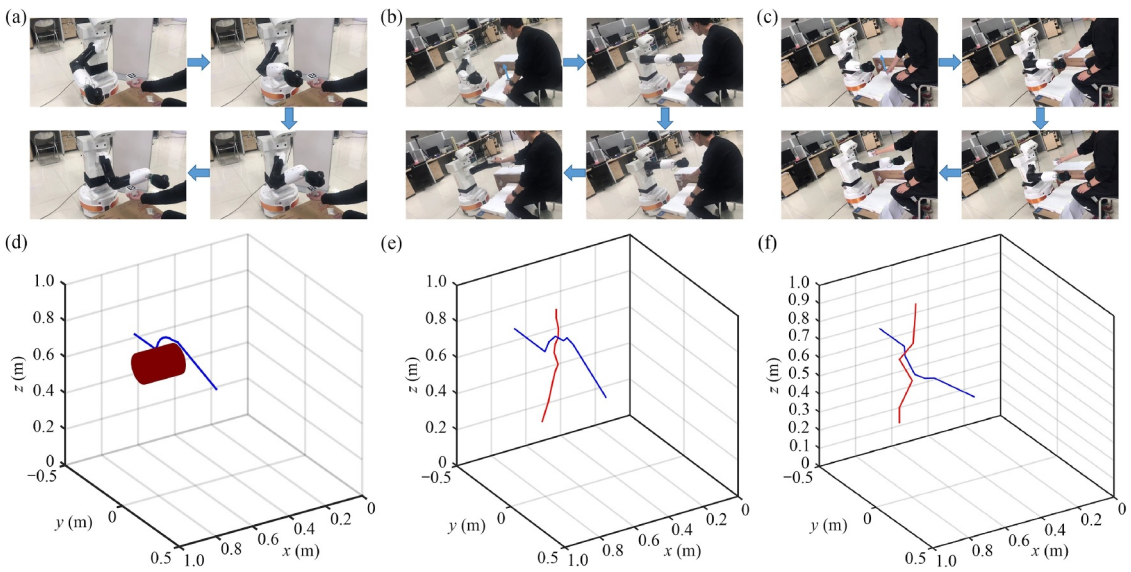


FIGURE 12 | Experimental results of utilising the proposed SDAPF algorithm to respectively avoid a static human hand, a slow-moving human hand and a fast-moving human hand are presented as follows: snapshots of the collision avoidance process are depicted in panels (a–c). The improved path (blue) of the manipulator to avoid the human hand (red) in a 3D view is shown in panel (d). Trajectories of the end effector (blue) and obstacles (red) are illustrated in 3D view in panels (e and f), respectively. As expected, in panel (e), the robot opts to maneuvre ahead of the human arm. Conversely, in panel (f), the robot prudently chooses to navigate behind the human hand, ensuring enhanced safety. This sequence of images provides a comprehensive depiction of the SDAPF algorithm's efficacy in distinct scenarios of human hand collision avoidance.

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from conventional obstacle avoidance methods. Simultaneously, the motion step is subject to adaptive adjustments.

5 | Conclusion

In this paper, we develop an advanced APF methodology, integrating position sampling and a novel danger index for enhancing collision avoidance in human–robot interactions. Our approach is distinguished by the introduction of a multi-layered bidirectional LSTM network, specifically designed for dynamic obstacle motion prediction, particularly focusing on hand movements within collaborative environments.

We have innovatively applied the concept of environmental complexity and variable step sizes to refine motion efficiency. This adaptive strategy allows for extended movement steps when obstacles are distant, transitioning to smaller, more cautious steps as the robot approaches an obstacle, thereby balancing efficiency with safety.

The introduction of a velocity potential function, grounded in the relative speed and distance to obstacles, underpins a more nuanced and responsive obstacle avoidance strategy. This enables the robot to dynamically select its avoidance path in real-time, based on the ongoing motion dynamics of the obstacles encountered.

Our experimental validation, conducted on the TIAGo robot platform, underscores the practicality and effectiveness of our proposed solutions. Through a series of crossover tasks, we demonstrated the robot's capability to make informed decisions based on the motion status of human collaborators, detected through advanced visual sensing techniques.

In the future, we aim to enrich our danger index by incorporating additional human-centric factors to further enhance the safety and intuitiveness of human–robot collaboration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Because of (privacy, security or proprietary) considerations, some data may not be publicly accessible. No publicly archived datasets were used in this study.

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